

BHUMIREDDY THARUN REDDY

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SUMMARY

Computer Science sophomore at IIIT Sri City (9.3 CGPA) selected for the prestigious **Amazon ML Summer School** cohort out of 130k+ applicants. Winner of the national **Syrotech Hackathon 2025** competing against 250+ teams. Deploys cross-domain expertise spanning production-ready full-stack backend development (NestJS, PostgreSQL) and advanced neural network architectures (PyTorch, Deep Learning).

EDUCATION

Indian Institute of Information Technology, Sri City

Andhra Pradesh, India

Bachelor of Technology – B.Tech, Computer Science (CGPA: 9.3 / 10)

Aug 2024 – Jul 2028

- **Core Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Probability & Statistics, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, Python, Java, SQL

Backend & Databases: Node.js, NestJS, RESTful APIs, PostgreSQL, MySQL, Supabase, MVC Architecture, Database Locking

AI / ML: PyTorch, MNE-Python, NumPy, Pandas, EEGNet, ShallowConvNet, OpenAI API, LangChain, SciPy, Sklearn

Tools & Software: Git, Docker, Streamlit, Postman, Linux, Jira, Vercel, Systems Design

HONORS & ACHIEVEMENTS

- **Selected Cohort Member, Amazon ML Summer School (2026):** Successfully cleared a highly competitive national selection test covering advanced Data Structures, Linear Algebra, and Probability; ranked among the top cohorts nationwide out of **130,000+ applicants**.
- **1st Place Winner, Syrotech Hackathon (2025):** Won the top position out of **250+ competing teams** nationwide for architecting an intelligent, highly scalable LLM-driven automated grading and educational platform.

PROJECTS

Restaurant Capacity & Lifecycle Management System | NestJS, TypeScript, PostgreSQL, REST APIs

Oct 2025

- Architected a scalable full-stack capacity management engine using NestJS and MVC design patterns, supporting complex multi-role workflows for customers, operational staff, and managers.
- Engineered live table seating and allocation tracking algorithms integrating state-driven reservation lifecycles, walk-in management, customer arrival check-ins, and automated grace periods to enforce no-show policies.
- Optimized PostgreSQL relational schemas and query execution paths with transaction-safe design patterns to handle heavy peak-hour traffic and eliminate concurrent double-booking race conditions.

IntelliGrader – AI-Powered Smart Education Platform | OpenAI API, Streamlit, Supabase PostgreSQL, YouTube API

Jan 2025

- Engineered an automated grading pipeline using OpenAI LLMs that evaluates complex answer scripts with high accuracy, generating granular, context-aware student feedback with or without explicit reference answer keys.
- Implemented an adaptive student workspace featuring a PDF semantic summarizer, a 24/7 interactive chat tutor, and automated extraction of weak learning points linked dynamically via the YouTube Data API for targeted recommendations.

EEG-Based Cognitive Stress & Motor Imagery Classification | PyTorch, MNE-Python, SciPy, Scikit-learn

Aug 2025 – Nov 2025

- Built an end-to-end BCI pipeline implementing a 4th-order Butterworth filter and FastICA via MNE-Python to dynamically isolate and eliminate EOG eye-blinks and muscle artifacts from EEG data (SAM40 and Dataset 2a).
- Deployed EEGNet-8,2 and ShallowConvNet architectures in PyTorch utilizing Mixup data augmentation, label smoothing, and Stratified K-Fold cross-validation to maximize subject-independent generalization.

EXPERIENCE & LEADERSHIP

Teaching Assistant (Data Structures & Algorithms and Signals & Systems)

Jan 2026 – May 2026

IIIT Sri City

Andhra Pradesh, India

- Mentored and evaluated a class of **100+ undergraduate students** on algorithmic concepts, computational complexity analysis, dynamic programming strategies, and optimized graph traversal implementations.
- Bridged theoretical Signals & Systems paradigms with optimized practical code implementations in core laboratory frameworks.

NLP Core Member

Aug 2025 – Apr 2026

Epoch – AI/ML Club, IIIT Sri City

Andhra Pradesh, India

- Collaborated on internal NLP sprints and orchestrated hands-on knowledge-sharing sessions centered on transformer workflows.

Web & Tech Core Member

Oct 2025 – Apr 2026

TEDx Sri City

Andhra Pradesh, India

- Maintained high-availability technical web infrastructure and monitored digital pipelines supporting large-scale event logistics.